

BankEase – Database Schema Design

Entity overview, table definitions, relationships, and key design notes.

1. Entity Overview

Table	Purpose
roles	Lookup of user roles (Customer, Employee, Administrator)
branches	Physical bank branches
employees	Bank staff, linked to the branch they work at
customers	Personal, identity and KYC details of bank customers (CIF, Bangla name, parents' and spouse's names, nationality, profession, income, source of fund, TIN, KYC status and verifier), registered at a branch
addresses	Present and permanent addresses of a customer (road/village, post office, thana, district)
users	Login credentials and role for a customer or employee (self-service or staff access)
account_types	Account types (e.g. Savings, Current) and their standard terms
accounts	A customer's individual bank account, with its account title
transactions	Deposits, withdrawals, and transfer legs posted to an account
transfers	Pairs a debit transaction with its matching credit transaction for a fund transfer
beneficiaries	A customer's saved transfer recipients
cards	Debit/credit cards issued against a customer's account
loans	A customer's loan application and its terms
loan_payments	Scheduled and paid installments against a loan
notifications	System notifications sent to a user
audit_logs	Record of significant actions taken in the system

2. Table Definitions

2.1 roles

Column	Type	Constraints
role_id	INT	PK, AUTO_INCREMENT
role_name	VARCHAR(50)	NOT NULL, UNIQUE — e.g. Customer, Employee, Administrator
description	VARCHAR(255)	NULL

2.2 branches

Column	Type	Constraints
branch_id	INT	PK, AUTO_INCREMENT
branch_name	VARCHAR(100)	NOT NULL, UNIQUE
address	VARCHAR(255)	NOT NULL
phone	VARCHAR(20)	NOT NULL
manager_employee_id	INT	FK → employees.employee_id, NULL (added after employees exists; see §3 Key Design Notes), ON DELETE SET NULL
opening_date	DATE	NOT NULL
status	ENUM('active','closed','under_renovation')	NOT NULL, DEFAULT 'active'

2.3 employees

Column	Type	Constraints
employee_id	INT	PK, AUTO_INCREMENT
full_name	VARCHAR(100)	NOT NULL
phone	VARCHAR(20)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
position	VARCHAR(100)	NOT NULL
branch_id	INT	FK → branches.branch_id, NOT NULL
hire_date	DATE	NOT NULL
employment_status	ENUM('active','on_leave','terminated')	NOT NULL, DEFAULT 'active'

2.4 customers

Column	Type	Constraints
customer_id	INT	PK, AUTO_INCREMENT
cif_number	VARCHAR(20)	NOT NULL, UNIQUE — bank customer information file number (CIF on the KYC form)
full_name	VARCHAR(100)	NOT NULL
full_name_bn	VARCHAR(100)	NULL — name in Bangla
father_name	VARCHAR(100)	NOT NULL
mother_name	VARCHAR(100)	NOT NULL
spouse_name	VARCHAR(100)	NULL — husband's / wife's name
date_of_birth	DATE	NOT NULL
gender	ENUM('male','female','other')	NOT NULL
nationality	VARCHAR(50)	NOT NULL, DEFAULT 'Bangladeshi'
profession	VARCHAR(100)	NOT NULL
monthly_income	DECIMAL(12,2)	NULL, CHECK (monthly_income IS NULL OR monthly_income >= 0)
source_of_fund	VARCHAR(255)	NOT NULL
tin_number	VARCHAR(20)	NULL, UNIQUE — if available
phone	VARCHAR(20)	NOT NULL, UNIQUE
email	VARCHAR(100)	NULL, UNIQUE
id_type	ENUM('nid','passport','birth_certificate','driving_license','other')	NOT NULL
id_number	VARCHAR(50)	NOT NULL, UNIQUE
branch_id	INT	FK → branches.branch_id, NOT NULL — registers_at
status	ENUM('active','inactive','blocked')	NOT NULL, DEFAULT 'active'
kyc_status	ENUM('pending','verified','expired','rejected')	NOT NULL, DEFAULT 'pending'
kyc_verified_by_employee_id	INT	FK → employees.employee_id, NULL — no ON DELETE action (MySQL does not allow one on a column used in a CHECK)
kyc_verified_at	DATETIME	NULL
registration_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
—	—	CHECK (kyc_status = 'pending' OR (kyc_verified_by_employee_id IS NOT NULL AND kyc_verified_at IS NOT NULL)) — any customer whose KYC has been looked at records who did it and when

2.5 addresses

Column	Type	Constraints
address_id	INT	PK, AUTO_INCREMENT
customer_id	INT	FK → customers.customer_id, NOT NULL, ON DELETE CASCADE
address_type	ENUM('present','permanent')	NOT NULL, DEFAULT 'present'
address_line	VARCHAR(255)	NOT NULL — road / village
post_office	VARCHAR(100)	NULL
thana	VARCHAR(100)	NULL
city	VARCHAR(100)	NOT NULL
district	VARCHAR(100)	NOT NULL
postal_code	VARCHAR(20)	NULL
country	VARCHAR(100)	NOT NULL, DEFAULT 'Bangladesh'
is_primary	BOOLEAN	NOT NULL, DEFAULT FALSE

2.6 users

Column	Type	Constraints
user_id	INT	PK, AUTO_INCREMENT
email	VARCHAR(100)	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	NOT NULL
role_id	INT	FK → roles.role_id, NOT NULL
linked_customer_id	INT	FK → customers.customer_id, NULL, UNIQUE (enforces 1:1)
linked_employee_id	INT	FK → employees.employee_id, NULL, UNIQUE (enforces 1:1)
failed_login_attempts	INT	NOT NULL, DEFAULT 0
status	ENUM('active','locked','disabled')	NOT NULL, DEFAULT 'active'
last_login_at	DATETIME	NULL
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
—	—	CHECK (linked_customer_id IS NOT NULL OR linked_employee_id IS NOT NULL) — every login maps to a customer or an employee

2.7 account_types

Column	Type	Constraints
account_type_id	INT	PK, AUTO_INCREMENT
type_name	VARCHAR(50)	NOT NULL, UNIQUE — e.g. Savings, Current
minimum_balance	DECIMAL(12,2)	NOT NULL, DEFAULT 0, CHECK (minimum_balance >= 0)
interest_rate	DECIMAL(5,2)	NOT NULL, DEFAULT 0, CHECK (interest_rate >= 0)
single_per_customer	BOOLEAN	NOT NULL, DEFAULT FALSE — whether a customer may hold only one account of this type

2.8 accounts

Column	Type	Constraints
account_id	INT	PK, AUTO_INCREMENT
account_number	VARCHAR(20)	NOT NULL, UNIQUE
account_title	VARCHAR(150)	NOT NULL — account holder's name as printed on the KYC form
customer_id	INT	FK → customers.customer_id, NOT NULL
account_type_id	INT	FK → account_types.account_type_id, NOT NULL
branch_id	INT	FK → branches.branch_id, NOT NULL
balance	DECIMAL(14,2)	NOT NULL, DEFAULT 0, CHECK (balance >= 0)
currency	VARCHAR(10)	NOT NULL, DEFAULT 'BDT'
status	ENUM('active','frozen','closed','pending')	NOT NULL, DEFAULT 'pending'
opening_date	DATE	NOT NULL
closing_date	DATE	NULL, CHECK (closing_date IS NULL OR closing_date >= opening_date)

2.9 transactions

Column	Type	Constraints
transaction_id	INT	PK, AUTO_INCREMENT
reference_number	VARCHAR(30)	NOT NULL, UNIQUE
account_id	INT	FK → accounts.account_id, NOT NULL
transaction_type	ENUM('deposit','withdrawal','transfer_in','transfer_out','fee','interest')	NOT NULL
amount	DECIMAL(14,2)	NOT NULL, CHECK (amount > 0)
balance_before	DECIMAL(14,2)	NOT NULL
balance_after	DECIMAL(14,2)	NOT NULL
related_account_id	INT	FK → accounts.account_id, NULL — counterparty account for transfers
description	VARCHAR(255)	NULL
status	ENUM('pending','completed','failed','reversed')	NOT NULL, DEFAULT 'completed'
performed_by_user_id	INT	FK → users.user_id, NOT NULL
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

2.10 transfers

Column	Type	Constraints
transfer_id	INT	PK, AUTO_INCREMENT
debit_transaction_id	INT	FK → transactions.transaction_id, NOT NULL, UNIQUE
credit_transaction_id	INT	FK → transactions.transaction_id, NOT NULL, UNIQUE
transfer_type	ENUM('internal','interbank')	NOT NULL, DEFAULT 'internal'
fee	DECIMAL(10,2)	NOT NULL, DEFAULT 0, CHECK (fee >= 0)
status	ENUM('pending','completed','failed')	NOT NULL, DEFAULT 'completed'

2.11 beneficiaries

Column	Type	Constraints
beneficiary_id	INT	PK, AUTO_INCREMENT
customer_id	INT	FK → customers.customer_id, NOT NULL, ON DELETE CASCADE
nickname	VARCHAR(50)	NULL
target_account_id	INT	FK → accounts.account_id, NOT NULL
added_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
—	—	UNIQUE (customer_id, target_account_id)

2.12 cards

Column	Type	Constraints
card_id	INT	PK, AUTO_INCREMENT
customer_id	INT	FK → customers.customer_id, NOT NULL
account_id	INT	FK → accounts.account_id, NOT NULL
card_type	ENUM('debit','credit')	NOT NULL
masked_number	VARCHAR(20)	NOT NULL — e.g. **** * 1234
card_number_hash	VARCHAR(255)	NOT NULL, UNIQUE
credit_limit	DECIMAL(12,2)	NULL, CHECK (credit_limit IS NULL OR credit_limit >= 0)
issue_date	DATE	NOT NULL
expiry_date	DATE	NOT NULL, CHECK (expiry_date > issue_date)
status	ENUM('active','blocked','expired','cancelled')	NOT NULL, DEFAULT 'active'

2.13 loans

Column	Type	Constraints
loan_id	INT	PK, AUTO_INCREMENT
customer_id	INT	FK → customers.customer_id, NOT NULL
loan_type	VARCHAR(50)	NOT NULL
principal_amount	DECIMAL(14,2)	NOT NULL, CHECK (principal_amount > 0)
interest_rate	DECIMAL(5,2)	NOT NULL, CHECK (interest_rate >= 0)
duration_months	INT	NOT NULL, CHECK (duration_months > 0)
application_date	DATE	NOT NULL
approval_date	DATE	NULL
disbursement_account_id	INT	FK → accounts.account_id, NULL
monthly_installment	DECIMAL(12,2)	NOT NULL
outstanding_balance	DECIMAL(14,2)	NOT NULL
status	ENUM('pending','approved','rejected','active','closed','defaulted')	NOT NULL, DEFAULT 'pending'
approved_by_user_id	INT	FK → users.user_id, NULL

2.14 loan_payments

Column	Type	Constraints
payment_id	INT	PK, AUTO_INCREMENT
loan_id	INT	FK → loans.loan_id, NOT NULL, ON DELETE CASCADE
installment_number	INT	NOT NULL, CHECK (installment_number > 0)
due_date	DATE	NOT NULL
scheduled_principal	DECIMAL(12,2)	NOT NULL
scheduled_interest	DECIMAL(12,2)	NOT NULL
paid_amount	DECIMAL(12,2)	NOT NULL, DEFAULT 0
paid_date	DATE	NULL
status	ENUM('due','paid','overdue','partial')	NOT NULL, DEFAULT 'due'
related_transaction_id	INT	FK → transactions.transaction_id, NULL
—	—	UNIQUE (loan_id, installment_number)

2.15 notifications

Column	Type	Constraints
notification_id	INT	PK, AUTO_INCREMENT
user_id	INT	FK → users.user_id, NOT NULL, ON DELETE CASCADE
type	VARCHAR(50)	NOT NULL
message	VARCHAR(255)	NOT NULL
is_read	BOOLEAN	NOT NULL, DEFAULT FALSE
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

2.16 audit_logs

Column	Type	Constraints
audit_id	INT	PK, AUTO_INCREMENT
user_id	INT	FK → users.user_id, NULL, ON DELETE SET NULL
action	VARCHAR(100)	NOT NULL
target_table	VARCHAR(50)	NOT NULL
target_id	INT	NULL
description	VARCHAR(255)	NULL
ip_address	VARCHAR(45)	NULL
result	ENUM('success','failure')	NOT NULL
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

3. Key Design Notes

- **One-to-one links:** users.linked_customer_id and users.linked_employee_id are each UNIQUE foreign keys, so a login maps to exactly one customer (self-service) or one employee (staff), never both at once being required — enforced by the CHECK on the users table.
- **Circular reference:** branches.manager_employee_id references employees, while employees.branch_id references branches. Create branches first with manager_employee_id left NULL, then create employees, then ALTER TABLE branches to add the manager_employee_id foreign key (or add it as a nullable FK from the start and populate it after both tables are seeded).
- **KYC data:** the customer and address fields follow the BRAC Bank “KYC Form for Beneficial Owner”: CIF, Bangla name, father’s / mother’s / spouse’s name, nationality, profession, monthly income, source of fund, TIN, ID (NID, passport, birth certificate or other), and separate present and permanent addresses with post office and thana. The account title is stored on accounts.
- **KYC verification:** customers.kyc_status starts as pending. Once it is anything else, a CHECK constraint requires the verifying employee (kyc_verified_by_employee_id) and the verification time to be recorded, so every KYC decision is traceable to a staff member.
- **Money fields** use DECIMAL(14,2) for account/transaction/loan balances and DECIMAL(12,2) for smaller fixed amounts (fees, installments, credit limits) to avoid floating-point rounding errors.
- **transactions** stores balance_before and balance_after as an audit-friendly ledger snapshot rather than recomputing balances from history on every read.
- **transfers** links a debit transaction to its matching credit transaction (1:1 on each side) instead of duplicating the transfer amount in a separate field.
- **cards.card_number_hash** stores a hash of the full card number; masked_number stores only the display-safe form (e.g. last 4 digits) so the full number is never stored in plaintext.
- **Indexes:** idx_transactions_account_date (account_id, created_at), idx_transactions_created, idx_customers_kyc_status, idx_loans_status, idx_loanpay_due (status, due_date), idx_notifications_unread (user_id, is_read) and idx_audit_created speed up the common lookups and reports.
- **Reporting views:** v_daily_transaction_summary (transactions per day), v_account_status_summary (accounts and balances per branch, type and status), v_cash_activity (deposits and withdrawals per branch per day) and

v_customer_kyc_overview (each customer's KYC status, verifier and verification time) support the management reports requested by the client.